

Git and GitHub

For the following questions, consider the file system below. You begin working in a shell at the Home directory, `~`.

```
~  
├─ practice  
│  ├── ex1.R  
│  └── ex2.R  
└─ my-project  
   ├── README.md  
   └── scripts  
       ├── data-cleaning.R  
       └── modeling.R
```

1. If you run the `ls` command, what will the output be?
2. What commands would you run to move into `my-project` and designate that directory as a git repository?
3. When you designate an existing directory as a git repo and it already has files in it, they will be neither staged nor committed. What single command could you use to stage the existing three files? Note that you can add multiple files at once by separating them by a space (see the wedding example)¹.
4. Once all of the files are staged, write the command can be used to take a snapshot of the state of those files along with a helpful caption.
5. Say that the project has a Part 1 and a Part 2. These existing files accomplish Part 1. To accomplish Part 2, you add several dozen lines of new code to `data-cleaning.R` and write and save a new script called `visualization.R` right alongside it. What commands would you run to store those changes along with a helpful description?

¹There are several other ways to do this. Feel free to check online for different methods.

Data Wrangling II

For this problem set, you'll be working with the `storms` tibble built into `dplyr`. This is a larger data frame than previous, so you may need to inspect it more carefully in R before proceeding with these questions. Please answer the question in common english and provide the code needed to produce the output used in your answer.

On the *challenging* exercises, a helpful approach is to start by sketching the data frame that would answer the question. What are the dimensions? What does each row refer to? Which columns are needed? Then piece together your pipeline to lead from `storms` to that data frame.

```
library(dplyr)
storms
```

```
# A tibble: 20,778 × 13
  name   year month   day hour   lat   long status   category   wind pressure
<chr> <dbl> <dbl> <int> <dbl> <dbl> <dbl> <fct>      <dbl> <int>    <int>
1 Amy   1975     6    27     0  27.5 -79 tropical d...    NA     25    1013
2 Amy   1975     6    27     6  28.5 -79 tropical d...    NA     25    1013
3 Amy   1975     6    27    12  29.5 -79 tropical d...    NA     25    1013
4 Amy   1975     6    27    18  30.5 -79 tropical d...    NA     25    1013
5 Amy   1975     6    28     0  31.5 -78.8 tropical d...    NA     25    1012
6 Amy   1975     6    28     6  32.4 -78.7 tropical d...    NA     25    1012
7 Amy   1975     6    28    12  33.3 -78 tropical d...    NA     25    1011
8 Amy   1975     6    28    18  34 -77 tropical d...    NA     30    1006
9 Amy   1975     6    29     0  34.4 -75.8 tropical s...    NA     35    1004
10 Amy  1975     6    29     6  34 -74.8 tropical s...    NA     40    1002
# i 20,768 more rows
# i 2 more variables: tropicalstorm_force_diameter <int>,
#   hurricane_force_diameter <int>
```

6. What are the dimensions of the data frame? What does each row correspond to (i.e. what is the unit of observation)?

7. Calculate the average wind speed and pressure for each category of storm.

8. Find the number of observations and the maximum wind speed for each combination of storm status and category. Sort the results by maximum wind speed in descending order².
9. Use the `.by` shortcut to calculate the average pressure for each year. Repeat the calculation using `group_by()`. Do you get the same answer?
10. For each storm name, calculate the z-score of wind speed (standardized within each storm) and store it in a new column called `wind_z`. Return the data frame that includes this new column.
11. *Challenging*. For each storm category, arrange the storms in descending order by their maximum wind speed. Retain only the top three storms in each category³.

²Hint 1: `n()` is a function that returns the row count when used inside `summarize()`. Hint 2: You can group by more than one variable.

³Hint: If you group by multiple variables, a call to `summarize()` will peel off the outermost grouping (the last argument in `group_by()`) and keep the others

12. *Challenging*: Create a summary that shows, for each year: the number of unique storms, the average duration (in hours) of storms, and the proportion of observations that are category 4 or 5 storms.

13. You may have noticed that your results for the answers in this problem set contain a value of `NA`. By default, most statistical summaries in R (`mean()`, `max()`, etc.) will return `NA` if any of the values are `NA`. If you prefer, you can calculate the mean after dropping the `NA` observations. One way to do this is to use the `na.rm = TRUE` option for functions that allow it (see `?mean`). The other way is to use the `drop_na()` function in `dplyr` (see `?drop_na()`).

Revisit your code above and, if you haven't already, drop the `NA` values in a column before calculating summary statistics.

More questions coming soon . . .